INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

TRACTOR, WHEELED, WAREHOUSE, INTERNAL COMBUSTION ENGINE, PNEUMATIC-TIRED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a heavy duty, internal combustion powered pneumatic-tired, industrial warehouse type tractor.

1.2 Classification. Warehouse tractors shall be one of the following sizes (see 6.2).

Size I - 4000 pounds (17,800 Newtons) drawbar pull.
Size II - 7500 pounds (33,375 Newtons) drawbar pull.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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SPECIFICATIONS

FEDERAL

GG-P-455	-	Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
VV-F-800	-	Fuel Oil, Diesel.

MILITARY

MIL-P-514	-	Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-L-2104	-	Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	-	Lubricating Oil, Gear, Multipurpose. (Metric)
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-S-43926	-	Suit, Chemical Protective.
MIL-G-43976	-	Glove and Glove Set, Chemical Protective.
MIL-F-43987	-	Footwear Cover, Chemical Protective (Overboots).
MIL-A-46153	-	Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	-	Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	-	Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	-	Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative. (Metric)
MIL-M-51282	-	Mask, Chemical Biological Field, M17A.
MIL-H-51291	-	Hood, Chemical Biological Mask M6A2.
MIL-C-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-V-81940	-	Valve, Sampling and Bleed, Hydraulic, Type II Systems.
MIL-T-83133	-	Turbine Fuel, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-162	-	Materials Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Process.
MIL-STD-209	-	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

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MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1474	-	Noise Limits for Army Materiel.
MIL-STD-1791	-	Designing for Internal Aerial Delivery In Fixed Wing Aircraft.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DOL)

CFR Title 29 - Chapter XVII Part 1910, Occupational Safety and Health Standards.

(Application for copies should be addressed to a local DOL office or The Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

MILITARY TRAFFIC MANAGEMENT COMMAND (MTMC/TEA)

Pamphlet 70-1 - Transportability for Better Strategic Mobility.

(Application for copies should be addressed to: The Military Traffic Management Command Transportation Engineering Agency (MTMC/TEA), ATTN: MTT-TR, P.O. Box 6276, Newport News, VA 23060-0276.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME/ANSI B56.9 - Safety Standard for Operator Controlled Industrial Tow Tractors.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

SAE J154	-	Operator Enclosures Human Factor Design Considerations.
SAE J209	-	Instrument Face Design and Location for Construction Industrial Equipment.
SAE J297	-	Operator Controls on Industrial Equipment.
SAE J534	-	Lubrication Fittings.
SAE J537	-	Storage Batteries.
SAE J551	-	Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
SAE J614	-	Engine and Transmission Dipstick Marking.
SAE J683	-	Tire Chain Clearance - Trucks, Buses (Except Suburban, Intercity, and Transit Buses), and Combinations of Vehicles.
SAE J753	-	Lubrication Chart - Construction and Industrial Machinery.
SAE J821	-	Electrical Wiring Systems for Construction, Agricultural and Off-Road Machines.
SAE J833	-	Human Physical Dimensions.
SAE J898	-	Control Locations for Off-Road Work Machines.
SAE J925	-	Minimum Service Access Dimensions for Off-Road Machines.
SAE J1127	-	Battery Cable.
SAE J1362	-	Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

Tire and Rim Association Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES, INC. (UL)

ANSI/UL 558	-	Standard for Industrial Trucks, Internal Combustion Engine - Powered.

(Application for copies should be addressed to Underwriters Laboratories, Publications Stock, 333 Pfingsten Road, Northbrook, IL 60062.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in and through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The industrial warehouse tractor, hereinafter called "tractor" shall be a heavy duty rear wheel drive, front wheel steer, pneumatic tired tow tractor having a minimum of 4000 lb or 7500 lb drawbar pull. The tractor shall have an internal combustion engine fueled by gasoline, diesel, or liquified petroleum gas.

3.1.1 Standard product. Except as otherwise specified herein, the tractor shall be the latest model of the standard commercial product of the manufacturer and shall be in production and in use by commercial users. The tractor shall be complete with all components that are standard with the manufacturer's products, whether stipulated herein or not, together with such accessories specified herein. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 First article. Unless otherwise specified (see 6.2), a sample (see 6.2) shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The tractor shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the tractor may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against electrolytic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of material and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

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3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the tractor may be newly fabricated from recovered materials to the maximum extent practicable, provided the tractor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the tractor.

3.3.3 Screw Threads. Screw threads shall be in accordance with FED-STD-H28.

3.3.4 Hazardous materials. In the event of products from foreign sources, care must be taken to ensure that lubricants and cooling oils in this item are free of polychlorinated biphenyls (PCB's).

3.3.5 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip coating compound or be furnished with a nonslip surface.

3.4 Operating temperature. The engine shall start without external preheating within 5 minutes and the tractor shall operate as specified herein within 15 minutes after engine start in any ambient temperature from -25 to +120 ^{0}F.

3.5 Performance. The tractor shall meet the performance requirements as specified herein.

3.5.1 Drawbar pull. The tractor shall be capable of developing and sustaining a drawbar pull at the coupler (see 3.12) of not less than 4000 pounds for Size I and 7500 pounds for Size II at elevations from sea level to 5000 ft (1524 meters) on a 0% slope.

3.5.2 Travel speed. The tractor shall be capable of attaining a maximum speed of no less than 12 miles per hour (19.31 Km/h), without a trailing load, in high gear on a level surface.

3.5.3 Right angle turn. The tractor shall be capable of turning in a 90 degree intersecting aisle whose width is no greater than 90 inches (2.29 meters) for Size I and 120 inches (3.05 meters) for Size II.

3.5.4 Endurance. The tractor, when tested in accordance with 4.7.2.4, shall be capable of completing 280 hours and a minimum of 1500 miles (2413 km) of endurance testing. The average cycle time shall not exceed 10 minutes for completion of a cycle as defined in 4.7.2.4.

3.6 Safety. Unless otherwise specified herein, the tractor shall conform to the requirements of UL 558, ASME/ANSI B56.9, and CFR Title 29 Part 1910 in effect at the time of manufacture and other requirements as specified. When specified (see 6.2) the tractor shall be gasoline safety (GS), diesel safety (DS), or liquified petroleum gas safety (LPS) rated in accordance with UL 558. All rotating or reciprocating parts and other parts subject to high

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operational temperatures that are of such a nature or are so located as to be a hazard to operating and maintenance personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Excessive projections of nuts and bolts, sharp edges, points and rough surfaces shall be eliminated except when required for normal operation of specified equipment.

3.6.1 Certification. Tractor shall bear UL stamp of approval as specified. In addition, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the tractor furnished under this specification meets other requirements of 3.6. Acceptable evidence of meeting these other requirements shall be certified test reports from recognized independent testing laboratories acceptable to the Government, indicating that the tractor conforms to the requirements of 3.6. Also the tractor shall be furnished with a safety nameplate specified herein (see 3.20).

3.7 Human factors engineering. The tractor shall be operable and maintainable by the 5th percentile female through 95th percentile male operators, in accordance with SAE J154 and SAE J833, who are wearing cold weather clothing, or when specified (see 6.2) wearing NBC protective gear (see 3.7.1). A weatherproof cushion seat and backrest shall be furnished on the tractor for the operator. Controls and displays shall be labeled with universal symbols in accordance with SAE J1362. Controls and operator seat shall be located in accordance with SAE J898 and SAE J297. Instrumentation shall be in accordance with SAE J209.

3.7.1 Nuclear-biological-chemical (NBC) protection. When specified (see 6.2), the tractor shall be operated by personnel wearing the following complete protection ensemble.

a. Mask, Chemical Biological, MIL-M-51282.
b. Hood, Chemical Biological, MIL-H-51291.
c. Suit, Chemical Protective, MIL-S-43926.
d. Glove Set, Chemical Protective, MIL-G-43976.
e. Footwear Cover, Chemical Protective, MIL-F-43987.

The operator shall be capable of actuating all controls (switches, pedals, knobs, and handles) without interference caused by the ensemble that is detrimental to the operation of the tractor. The protective ensemble as listed above, will be provided as Government-loaned property to the contractor in order to permit performance of the test.

3.8 Maintainability. Provisions shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, battery, belts, wear parts of steering assembly, tires, wheels, lights, and horn. Dimensions of access opening shall be in accordance with SAE J925 for mittened hand. There shall not be interference by frame members or other obstructions when servicing or drawing oils and lubricants to or from any assembly or component. Flexible connections shall not be required. All reservoirs with removable drain plugs shall provide for complete drainage into a container. Engine and transmission dipsticks shall be in accordance with SAE J614.

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3.8.1 Maintenance operations. Unless otherwise specified herein, each operation shown on the following maintenance operations list shall be accomplished by one man, using common tools, in not more than the time specified.

a. Remove, replace and adjust engine driven belts - 30 minutes.
b. Remove and replace alternator and adjust belt tension - 45 minutes.
c. Remove and replace fuel filter elements - 15 minutes.
d. Disconnect battery cables, remove and replace battery and reconnect battery cables - 30 minutes.
e. Drain engine crankcase, remove and replace all filter elements, strainers and refill - 45 minutes.
f. Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers and refill converter and transmission - 1 hour.
g. Remove and replace starter - 1 hour.
h. Bleed and adjust brakes and refill master cylinder (two men) - 45 minutes.
i. Remove and replace engine coolant system hoses - 30 minutes.
j. Remove and replace headlights and tail light bulb(s) - 15 minutes each bulb.
k. Remove, disassemble and reinstall air cleaner - 30 minutes.
l. Lubricate all lubrication fittings with 2 strokes per fitting using grease gun and extension - 30 minutes.
m. Reset circuit breaker or change fuse - 10 minutes.
n. Remove and replace all fluid level dipsticks - 10 minutes.

3.8.2 Lubrication. Lubrication means shall be provided for all moving parts that require lubrication, except where sealed permanent lubrication is provided. Each tractor shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-G-10924, and MIL-B-46176 as applicable. Initial servicing and run-in may be accomplished with lubricants standard to the contractor's commercial practice (except for braking system) if compatible with above military lubricants. (If commercial lubricants are utilized, the manufacturer will provide certification from suppliers that commercial lubricants are compatible with military lubricants prior to the start of first article test). A lubrication data plate in accordance with SAE J753 shall be mounted on the tractor and shall identify all military lubricants and appropriate temperature ranges to be utilized when servicing the tractor. The data plate shall conform to MIL-P-514, type II, composition C (GG-P-455), type I, grade A, class 1. Grease lubrication fittings shall conform to SAE J534, and shall be in a protected but accessible position.

3.9 Transportability. The tractors shall be capable of being transported by military or commercial trailers, trains, marine vessels, and aircraft, and must be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation.

3.9.1 Slinging and tiedown provisions. The slinging and tiedown provisions shall be provided and shall conform to MIL-STD-209, class 1 and 2 or 3 for

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type III equipment. The provisions shall enable the complete tractor to be lifted in the normal operating position. All slinging/tiedown provisions shall be labeled "LIFT", "TIEDOWN", or "LIFT/TIEDOWN" as appropriate (see 3.19).

3.9.2 Air transportability. The tractor shall meet the requirements of MIL-STD-1791 for air transport in U.S. Air Force C-130, C-141, and C-5A aircraft.

3.9.3 Rail transportability. The tractor shall be rail transportable in CONUS and NATO countries without restrictions. The tractor shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with MTMC/TEA pamphlet 70-1, outline diagram when loaded on a 50-inch (127 cm) high rail car. The tractor shall be capable of withstanding shock loads resulting from rail impact testing in accordance with MIL-STD-810 (see 4.7.5.3) without failure, damage, or permanent deformation.

3.9.4 Highway transportability. The tractor when loaded on a semitrailer/tractor, shall be within the highway permit limits of all States.

3.10 Engine. The tractor shall be furnished with a diesel, gasoline, or liquified petroleum gas internal combustion engine as specified in 6.2. The engine shall be covered by an engine cowl or hood which can be easily opened or removed without tools. The engine shall be capable of operating on the applicable military lubricants specified in 3.6.2. If a diesel engine is furnished, it shall operate on fuel conforming to both VV-F-800, grade DF2 and MIL-T-83133.

3.10.1 Fuel System. The capacity of the fuel storage tank shall ensure not less than 6 hours of continuous operation during first article endurance testing specified in 3.5.4. A fuel filter shall be provided and shall be accessible from above or from the side without removal of other components.

3.10.1.1 Gasoline or diesel engine fuel system. The fuel system shall include a water separator located in the fuel line before the fuel filter. The fuel tank shall be equipped with a safety filler cap assembly, painted with red enamel for a gasoline system and green enamel for a diesel system, and shall incorporate a self-closing, cap retaining device, removable strainer and means for padlocking. The fuel type shall be labeled near the fill port (see 3.19). The fuel tank shall be equipped with a drain plug which shall be removable with hand tools without having to remove any other component.

3.10.1.2 Liquified petroleum gas (LPG) engine fuel system. An automatic shut-off valve shall be provided in the fuel system to prevent flow of fuel when the ignition is off and the engine is not running or if the engine should stop.

3.10.2 Cold weather starting system. When a diesel engine is specified, integral starting aids shall be provided to enable the engine to be started under cold weather conditions down to -25 ^{0}F. If cold weather start system

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uses ether, then a measured shot system shall be employed that is not functional before cranking engine and not functional after the engine has started. LPG and gasoline engines shall start without starting aids.

3.10.3 Crankcase ventilation system. A positive crankcase ventilation system shall be furnished that shall prevent crankcase vapor from entering the atmosphere.

3.10.4 Alternator. A 12-volt alternator of not less than 40-ampere capacity shall be provided and shall provide charge at engine low idle speed and maintain battery charge at full electric load. An alternator regulator shall be furnished. The regulator may be integral with the alternator.

3.10.5 Starting system. A device shall be furnished to prevent re-energizing the starter motor after the engine has been started. Tractors shall be provided with an interlock in the starting system or other means to prevent energizing the starter motor except when the transmission is in the neutral or park position.

3.10.6 Air cleaner. A dry replaceable cartridge type air cleaner shall be furnished. The cleaner shall be a multistage air cleaner and shall include a dust scavenger. It shall be accessible without removal of other components when engine cowl or hood is open. A restriction indicator shall be furnished to indicate when the replaceable filter element shall be changed, and shall be visible from the operators seat.

3.10.7 Exhaust system. Mufflers and exhaust piping shall conform to the requirements of UL 558. If a vertical exhaust discharge is used, a rain cap shall be provided. Exhaust discharges shall be directed away from personnel.

3.10.8 Cooling system. When a liquid-cooled engine is furnished, the cooling system shall be of the closed pressure type, incorporating a radiator with a pressure cap, hoses, thermostat, fan, and coolant circulating pump. The cooling system fluid shall be a 50-50 mixture of antifreeze conforming to MIL-A-46153 and water. Accessible drain(s) shall be provided to completely drain the cooling system directly into containers.

3.10.9 Oil sampling valve. An oil sampling valve in accordance MIL-V-81940 shall be provided on the engine and transmission. The valve shall be located in such a position as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the valve shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. The sampling valve shall be labeled adjacent to the valve indicating type of oil sample (engine or transmission, whichever is appropriate).

3.11 Drive train components. The tractor shall be equipped with a torque converter, transmission, drive shaft with universal joints, front steer axle, rear drive axle, and wheel and tire assemblies. All rotating shafts and axles shall be supported on sealed or lubricated bearings. All gears shall operate in the appropriate lubricant specified in 3.8.2.

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3.11.1 Axles. The front and rear axles shall be suspended on semi-elliptical springs.

3.11.2 Transmission. The tractor shall be equipped with a full-torque automatic shift transmission having a minimum of 2 forward and 1 reverse speeds.

3.11.3 Wheels. Wheel arrangement for the tractor shall consist of two front steer wheels and two or four rear drive wheels. Steer wheels with new tires shall clear any part of the tractor structure by at least 3/8 inch (.95 cm). Tire chain clearance for drive wheels shall be not less than clearance specified in SAE J683.

3.11.4 Tires and tire loading. Tires shall be new and unused. Tire loadings shall not exceed the values listed in the Tire and Rim Association Year Book. Valve caps shall be of metal with elastomer inner seal. Tires shall be not more than 12 months old on date of acceptance by the Government. Tire pressure shall be stenciled on the tractor above each tire (see 3.19). No tire shall protrude beyond the plan outline of the tractor when the steer wheels are straight ahead.

3.12 Coupler. The tractor shall be equipped with a rear mounted semi-automatic pintle type coupler, mounted 12 inches (30.5 cm) ±1/2 inch (1.3 cm) above ground level with the tires at their recommended inflation pressures. The coupler shall consist of a coupler jaw configuration with a minimum throat of 1-3/4 inches (4.45 cm) and a minimum throat length of 5-3/4 inches (14.6 cm), and an accompanying spring loaded latch and pin.

3.13 Steering. The front steerable wheels on the tractor shall be controlled by a steering wheel. The steering wheel shall require less than 50 pounds (22.7 kg) of pull at the rim to turn the wheel when moving forward at 1 mph (1.61 km/h) with a dead engine.

3.14 Brakes.

3.14.1 Service brakes. All wheel service brakes shall be provided and shall be of the hydraulic internal expanding type or hydraulically actuated disc type. In the event the brake hydraulic system fails on one axle, the other axle shall have its normal braking capacity. The service brake system performance shall be in accordance with ASME/ANSI B56.9 for a speed of 12 mph (19.31 km/h).

3.14.1.1 Brake fluid. Brake system shall be compatible with and filled with MIL-B-46176 Brake Fluid.

3.14.2 Parking brake. A parking brake shall be required meeting the parking brake system performance requirements of ASME/ANSI B56.9 on a 5% slope. The parking brake shall provide braking to a minimum of two wheels either through the drive line or the wheels themselves. Not more than 30 pounds (13.7 kg) of pull force on the lever shall be required to apply the parking brake.

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3.15 Closed cab. When specified (see 6.2), tractors shall be equipped with a removable rigid metal closed cab equipped with two (easily removable) doors for use in inclement weather. The cab shall include windshield shatterproof glazing, one or two power operated windshield wipers, two side windows, and one rear window. Provisions shall be made for manual operation of windshield wipers in the event of power or mechanical failure. A rearview mirror providing maximum visibility to the rear of the tractor shall be mounted inside the cab in view of the operator. Cab glazing shall provide a seated operator all-around visibility. Provisions shall be made for ventilation. Minimum clearance shall be in accordance with SAE J154. An automotive type heater with windshield defroster attachments shall be furnished for the closed cab. When specified (see 6.2), the cab shall be equipped with a yellow rotating warning light which is operated by a switch mounted on the dashboard. The light shall be mounted on the top of the cab in a location which provides all-around visibility of the light.

3.16 Controls and instrumentation. Controls and instrumentation shall be located and labeled in accordance with 3.7. Electric controls and instrumentation shall be moisture and weather resistant to prevent entry of moisture when operated and stored out of doors under all weather conditions.

3.16.1 Ammeter gauge. A remote indicating ammeter gauge shall be furnished.

3.16.2 Engine oil pressure gauge. A remote indicating oil pressure gauge which is marked in red in the low pressure zone shall be furnished.

3.16.3 Fuel level gauge. An electric, thermally stabilized, remote indicating fuel gauge shall be provided on diesel and gasoline fueled tractors.

3.16.4 Engine coolant temperature gauge. For liquid cooled engines, a remote indicating engine coolant temperature gauge marked in red in the high temperature zone shall be furnished.

3.17 Electrical system. Electrical system voltage shall be 12 Vdc. Electrical wiring systems shall be in accordance with SAE J821.

3.17.1 Ignition switch. A key-operated ignition switch shall be furnished. Two ignition keys will be furnished with each tractor. All keys shall be identical for every tractor.

3.17.2 Headlights. The tractor shall be furnished with two headlights operated by a switch mounted on the dashboard.

3.17.3 Taillight. A minimum of one automotive type combination stoplight and taillight shall be furnished.

3.17.4 Backup light. A minimum of one backup light which automatically lights when the tractor is placed in reverse shall be furnished.

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3.17.5 Horn. One electric horn shall be provided.

3.17.6 Battery. Tractors shall be furnished with a minimum of one maintenance free 12-volt battery conforming to SAE J537.

3.17.6.1 Battery enclosure. A weather protected box or compartment type battery enclosure, shall be provided and be accessible for servicing without removal of the battery or other components except the quick release battery box (if weather protected box is furnished). The enclosure shall be equipped with battery hold down components to prevent any damage to the batteries. The enclosure shall include a drain for water or electrolyte, and a vent to prevent gas accumulation. Battery cables conforming to SAE J1127 shall be furnished with insulated terminal covers. Positive and negative cable terminals shall be identified with red sleeve, labeled "+" and a black sleeve, labeled "-", respectively and corrosion-resistant bolts and nuts provided. Negative ground shall be provided.

3.17.7 Hourmeter. An electrically operated hourmeter which registers the number of engine operating hours up to a minimum of 9999 hours shall be provided. The hourmeter shall be accessible for reading without removing any component of the tractor.

3.18 Treatment and painting. The parts of the tractor normally painted shall be cleaned, treated and painted in accordance with the contractor's normal commercial practice. When specified (see 6.2), treatment, priming, and top coating shall be performed in accordance with MIL-T-704, type A, top coat color yellow number 13538. When specified (see 6.2) chemical agent resistant coatings, (CARC) in accordance with MIL-C-46168 or MIL-C-53039 shall be used. When CARC is specified, treatment, priming and topcoating shall be done in accordance with MIL-T-704, type F or G as applicable, topcoat color green 383.

3.19 Tractor marking. Markings on the tractor shall be painted black, unless another color is specified (see 6.2). Characters shall be block-type capital letters and Arabic numerals. Location and character size shall be as shown in table I.

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TABLE I. Marking, location and size.

Information	Example	Location	Minimum Character Size (Ht.)
Agency	US Army	On each side & rear	1-1/2 inches (3.8 cm)
Registration No.	10D315	On each side & rear	1-1/2 inches (3.8 cm)
Capacity	4000 lb. DBP	On each side	2 inches (5.1 cm)
Safety	No Smoking	On side panel	1 inch (2.5 cm)
	No Riders	On side panel	1 inch (2.5 cm)
Tire pressure	TP100	On frame above each tire	1 inch (2.5 cm)
Type fuel		Gasoline Near fuel tank filler	1 inch (2.5 cm)
Lifting/Tiedown points	Lift/Tiedown	Near lift/tiedown points	1 inch (2.5 cm)

3.20 Identification marking. The tractor shall be identified with an identification plate conforming to MIL-P-514. The plate shall be located in a visible and permanent location on the instrument panel or firewall of the tractor. Also, state the independent testing laboratory name, independent laboratory registration or index number assigned to the inspected truck, and date and location of independent laboratory inspection. The identifying data to be applied to the identification plates shall be stamped in accordance with MIL-P-514. The plate shall be securely attached to the tractor with screws, bolts or rivets, and shall be furnished and mounted by the contractor.

3.21 Noise limits. Noise levels produced by the tractor shall comply with the requirements of MIL-STD-1474 except as indicated herein.

a. The noise level at the operator's position and at one meter from the perimeter of the tractor at all other locations shall not exceed Category D of table II (85 db(A)) of this specification. MIL-STD-1474, 5.1.1.2, 5.2, 5.3, and 5.4 shall not apply. Compliance and non-compliance with the required noise limits shall be documented based upon test requirements of 4.7.9 and the data requirements of MIL-STD-1474, 5.5.
b. The following table II shall be used in lieu of table II contained in MIL-STD-1474.

Table II. Steady-state noise limits (dB(A)) for personnel-occupied areas.

	Limit Category					
	A*	B*	C*	D*	E**	F**
A-weighted Limit [dB(A)]	108	100	90	85	75	65

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3.22 Electromagnetic radiation. The tractor shall not exceed the limits of electromagnetic radiation specified in SAE J551.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements examined and tested as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.6).
c. Inspection comparison (see 4.9).
d. Inspection of packaging (see 4.10).

4.3 Inspection conditions. Unless otherwise specified herein, all examinations and tests shall be performed at the ambient temperature and climatic conditions existing at the place of examination and test. Only that maintenance established by the contractor and submitted as the maintenance schedule (see 4.4.c) prior to commencement of the tests shall be performed during the tests.

4.4 First article inspection. The manufacturer shall furnish one or more tractors (see 6.2) produced from production tooling for first article evaluation and test (see 6.3). Prior to examination and test of the tractor specified in 4.4.1 and 4.4.2, the following shall be performed:

a. Service the tractor with oils, greases, and fuels specified herein and designated for use in the ambient temperatures at which the test will be conducted.

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b. Conduct a run-in by performing one cycle on the endurance test course as specified in 4.7.2.4.
c. Furnish a schedule of maintenance to be followed during all testing of the tractor. A list of tools required to perform this maintenance shall also be required.

4.4.1 First article examination. Prior to testing, the first article tractor(s) shall be examined for the defects marked "x" in column 1 of table III. Presence of one or more defects shall be cause for rejection.

4.4.2 First article tests. Upon successful completion of the examinations specified in 4.4.1 the tractor(s) shall be subjected to the tests marked "x" in column 1 of table IV. Acceptance of the first article tractor shall not exclude the remaining tractors from quality conformance inspection and acceptance provisions specified in section 4.

4.5 First article inspection failure. Failure of a first article or production tractor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 or 4.6 shall be cause for rejection of the first article tractor(s) and for refusal by the Government to continue acceptance of production tractors until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the manufacturer at no cost to the Government on tractors previously accepted and produced under the contract. Any deficiencies found as a result of the first article tractor inspection will be considered prima facie evidence that all tractors accepted prior to the completion of the first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Quality conformance inspection.

4.6.1 Quality conformance examination. After successful completion of the quality conformance tests specified in 4.6.2, each tractor shall be examined for the defects marked "x" in column 2 of table III. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance tests. Each tractor shall be subjected to the production run-in as specified in 4.8.

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TABLE III. Examination schedule.

First Article	Quality Conformance	Defect	Requirements Paragraph
1	2	3	4
X	-	101. Materials not as specified.	3.3
X	-	102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for applicable storage and operating environments.	3.3.1
X	-	103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
X	X	104. Contractor does not have documentation available for identification of material, material finishes, or treatments.	3.3.1.2
X	X	105. Used rebuilt or remanufactured components, pieces, or parts incorporated in the tractor.	3.3.2
X	-	106. Screw threads not as specified.	3.3.3
X	-	107. No PCB's are used.	3.3.4
X	X	108. Walkways not nonslip as specified.	3.3.5
X	X	109. Safety not as specified.	3.6
X	X	110. UL stamp of approval missing.	3.6.1
X	X	111. Human factors not as specified.	3.7
X	-	112. Maintainability provisions not as specified.	3.8
X	X	113. Lubrication provisions and lubricants not as specified.	3.8.2
X	X	114. Engine not as specified.	3.10
X	X	115. Fuel system not as specified.	3.10.1
X	X	116. Gasoline or Diesel fuel system not as specified.	3.10.1.1
X	X	117. LPG fuel system not as specified.	3.10.1.2
X	X	118. Cold weather starting aids not as specified.	3.10.2
X	-	119. Crankcase ventilation not as specified.	3.10.3
X	X	120. Alternator not as specified.	3.10.4
X	X	121. Starting system not as specified.	3.10.5
X	X	122. Air cleaner not as specified.	3.10.6
X	-	123. Exhaust system not as specified.	3.10.7
X	-	124. Cooling system not as specified.	3.10.8

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TABLE III. Examination schedule - continued.

First Article	Quality Conformance	Defect	Requirements Paragraph
1	2	3	4
X	X	125. Oil sampling valves and labels not as specified.	3.10.9
X	X	126. Drive train components not as specified.	3.11
X	X	127. Axles not suspended.	3.11.1
X	X	128. Transmission not as specified.	3.11.2
X	X	129. Wheels not as specified.	3.11.3
X	X	130. Tires and tire loadings not as specified.	3.11.4
X	X	131. Coupler not as specified.	3.12
X	-	132. Steering not as specified.	3.13
X	-	133. Service brakes not as specified.	3.14.1
X	X	134. Brake fluid not as specified.	3.14.1.1
X	-	135. Parking brake not as specified.	3.14.2
X	X	136. Cab and cab components not as specified.	3.15
X	X	137. Controls and instrumentation not as specified.	3.16
X	X	138. Ammeter gauge not as specified.	3.16.1
X	X	139. Engine oil pressure gauge not as specified.	3.16.2
X	X	140. Fuel level gauge not as specified.	3.16.3
X	X	141. Engine coolant temperature gauge not as specified.	3.16.4
X	-	142. Electrical system not as specified.	3.17
X	X	143. Ignition switch not as specified.	3.17.1
X	X	144. Headlights not as specified.	3.17.2
X	X	145. Taillight(s) not as specified.	3.17.3
X	X	146. Backup light(s) not as specified.	3.17.4
X	X	147. Horn not as specified.	3.17.5
X	X	148. Battery not as specified.	3.17.6
X	X	149. Battery enclosure not as specified.	3.17.6.1
X	X	150. Hourmeter not as specified.	3.17.7
X	X	151. Treatment and painting not as specified.	3.18
X	X	152. Markings not as specified.	3.19
X	X	153. Identification plate missing or not as specified.	3.20

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TABLE IV. Test schedule.

First Article	Test	Test Paragraph	Requirement Paragraph
1	2	3	4
X	Test conditions.	4.3	3.6, 3.7, & 3.8
X	Operating temperature.	4.7.1	3.4
X	Drawbar pull.	4.7.2.1	3.5.1
X	Travel speed.	4.7.2.2	3.5.2
X	Right angle turn.	4.7.2.3	3.5.3
X	Endurance.	4.7.2.4	3.5.4
X	Nuclear-Biological-Chemical.	4.7.3	3.7.1
X	Maintenance operations.	4.7.4	3.8.1
X	Slinging and tiedown.	4.7.5.1	3.9.1
X	Air transportability.	4.7.5.2	3.9.2
X	Rail impact.	4.7.5.3	3.9.3
X	Highway transport.	4.7.5.4	3.9.4
X	Starter disconnect.	4.7.6	3.10.5
X	Steering.	4.7.7	3.13
X	Braking.	4.7.8	3.14.1 & 3.14.2
X	Noise level measurement.	4.7.9	3.21
X	Electromagnetic radiation.	4.7.10	3.22

4.7 Tests.

4.7.1 Operating temperature. Demonstrate that the tractor can start without preheating within 5 minutes and operate as specified within 15 minutes after engine start as specified in 3.4. The tractor shall have been cold soaked to attain a stabilizing temperature specified. Nonconformance shall constitute failure of this test.

4.7.2 Performance.

4.7.2.1 Drawbar pull. Place the tractor on a level, smooth, concrete, or paved surface, with tires inflated to TRA recommended pressure. Attach a tension dynamometer attached to a suitable anchor and the tractor's pintle installed 12 inches (30.5 cm), ±1/2 inch (1.3 cm) above ground level. Operate tractor at maximum drawbar pull for at least 20 seconds and record drawbar pull at 5 second intervals. Repeat at least 5 times. Nonconformance to 3.5.1, any malfunction, overheating, leakage of fluid, or failure of any components shall be cause for rejection of the tractor by the Government.

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4.7.2.2 Travel speed. Operate the tractor on a level, smooth, concrete, or paved surface without a trailing load. Drive the tractor the distance necessary to attain maximum speed prior to entering a measured distance of not less than 22 feet (6.7 meters). Record time to traverse this distance; the tractor shall be driven over the measured course at least two times in each direction. The speed shall be the average of the four runs. Inability of the tractor to attain a maximum speed of not less than 12 mph (19.31 Km/h) in high gear (see 3.5.2), failure of any components, overheating, malfunction, or leakage of fluid shall be cause for rejection of the tractor by the Government.

4.7.2.3 Right angle turn. For a size I tractor, construct a course to simulate 90-inch (2.29 meters) wide aisles intersection at 90 degrees. For size II tractor, change dimensions to 120 inches (3.05 meters) for testing of tractor (see 3.5.3). Operate the tractor through this course in both directions in one continuous motion without backing. Tractor contact with any part of the simulated aisles shall be cause for rejection.

4.7.2.4 Endurance. A test course of a shape and size most feasible for the space available and which allows the tractor to be driven in a continuous forward direction while towing a trailing load shall be established. The test course shall be paved with concrete, asphalt, macadam, or equivalent and shall incorporate a ramp and obstacle course. One cycle on the test course shall consist of traveling 2640 feet (804.67 m) in a forward direction which includes traversing a ramp and obstacle course (except with trailing load) once each cycle. The ramp shall have a 5% slope (plus or minus 1 percent slope) and have the approximate dimensions and configuration as shown in figure 1. The obstacles shall be fixed to the test course and shall be configured as shown in figure 2. A minimum trailer and load combination of 10 times the drawbar rating for the size tractor tested (Example: Size I tractor with 4000 pound (1814.4 kg) drawbar pull shall have a minimum trailer and load combination weight of 40,000 pounds (18,144 kg)) shall be prepared for use in the test. The load shall be any desired material uniformly distributed over the load carrying surface of the trailer. Each cycle of the test shall begin by positioning the tractor at a predetermined starting point on the test course (any part of the course except for the ramp or obstacle course). All switches including the horn shall be operated. The headlights and warning light (if so equipped) shall be turned on. The tractor pulling the trailing load shall proceed in a forward direction at any desired speed. Approximately half-way up the 5% slope the tractor shall be brought to a stop using the service brakes, the transmission placed into neutral, the parking brake applied, and the service brake released. The parking brake shall be released and the tractor shall proceed on the course (without traversing the obstacle course) to the starting point. The tractor shall be stopped using the service brakes at the starting point and the headlights/warning light shall be turned off. The next cycle shall then be performed identical to the previous. At the end of one hour of operation, the engine shall be stopped and the trailing load disconnected from the tractor. The tractor shall perform each cycle the same as when pulling the trailing load except that the tractor shall transverse the obstacle course once during each cycle. At each successive one hour interval, the tractor shall be operated alternately with and without the

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specified trailing load. Inability to pull load, complete the operation, overheating, leakage of fluid, malfunction, failure of any component, or inability to meet requirement of 3.5.4 shall be cause for rejection of the tractor by the Government until such malfunction is correct. Any unscheduled maintenance action during this test taking one person more than 15 minutes to correct or the total time of malfunction exceeding 5 percent of the time required to complete the specified number of hours shall be considered failure of the endurance test specified in 3.5.4. During this test, the tractor shall operate for a minimum of 6 hours without refueling as specified in 3.10.1.

4.7.3 Nuclear-biological-chemical operations test. The tractor shall complete two cycles on the endurance course specified in 4.7.2.4 within 24 minutes with the operator wearing the complete protection ensemble specified in 3.7.1. All controls and switches shall be operated during each cycle. Inability to complete the cycles in the time specified or inability of the operator to operate all controls as specified in 3.7.1 shall constitute failure of this test.

4.7.4 Maintenance operations. Perform the maintenance operations as listed in 3.8.1. Nonconformance to 3.8.1 shall constitute failure of this test.

4.7.5 Transportability.

4.7.5.1 Slinging and tiedown provisions test. The slinging and tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.9.1. Inability to meet the requirements of 3.9.1 shall constitute failure of this test.

4.7.5.2 Air transportability. The tractor shall be tested in accordance with MIL-STD-1791 to prove conformance to 3.9.2. Inability to meet the requirements of 3.9.2 shall constitute failure of this demonstration.

4.7.5.3 Rail impact test. The first article tractor shall be tested in accordance with MIL-STD-810 to prove conformance to 3.9.3. Inability to meet the requirements of 3.9.3 shall constitute failure of this test.

4.7.5.4 Highway transport test. The tractor shall be tested to conform to 3.9.4. Inability to meet the requirements of 3.9.4 shall constitute failure of this test.

4.7.6 Starter disconnect. Start the tractor, energizing the starter switch. Allow the engine to run for a minimum of 15 seconds. Re-energize the starter switch while the engine is running. Any evidence of starter engagement while engine is running shall be cause for rejection (see 3.10.5).

4.7.7 Steering test. With the engine stopped and the tractor traveling in the forward direction at a speed of 1 mph (1.61 Km/h) ± 1/2 mph (0.81 Km/h) measure the force tangential to the steering wheel required to turn the rim of the steering wheel. Nonconformance to 3.13 shall constitute failure of this test.

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4.7.8 Braking test. Perform applicable braking tests specified in 3.14.1 and 3.14.2. Nonconformance shall be cause for rejection.

4.7.9 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal arc increments, one increment shall include data from the noisiest position. Additionally the noise level at the typical operating position shall be provided as dB(A) level.

4.7.10 Electromagnetic radiation. Test the tractor in accordance with SAE J551. As a minimum, the electromagnetic radiation report shall include results of all testing and a list of components used to reduce the electromagnetic radiation. Measurements shall also be taken from the front and rear of the tractor at 33 foot (10 meter) distances. Nonconformance to 3.22, shall constitute failure of this test.

4.8 Production run-in. Each tractor shall be subjected to a production run-in and servicing as specified herein. Inability to complete run-in, overheating, failure of any components, malfunction of any control, leakage of fluids shall be cause for rejection of the tractor by the Government.

4.8.1 Drive train. Apply the service brakes, shift the transmission into gear, and apply wide open throttle for not less than 5 seconds. Repeat in all gears including reverse.

4.8.2 Travel. Start tractor and operate through all gears including reverse and then travel in high gear for a distance of at least 1 mile (1610 meters). Operate all lights, horn, and brake system not less than 2 times during this demonstration.

4.8.3 Service and adjustment. Prior to delivery, the manufacturer shall service and adjust each tractor for customer operational use. The service and adjustment will include, at a minimum, the following:

a. Adjust steering and align front end assembly.
b. Adjustments of engine, transmission and power train.
c. Adjustment of brake system.
d. Check electrical systems continuity, fill and charge the battery.
e. Check torque on all wheel lug nuts and proper inflation of tires.
f. Service the cooling system with an antifreeze solution as specified in 5.1.
g. Check fluid levels and fill to required level if necessary.

4.9 Inspection comparison. The Government may select tractors at any time during the contract production period and subject the tractors to the examination specified in column 1 of table III and to the tests specified in column 1 of table IV to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have

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been accepted by the Government and will not include the previously inspected first article tractors. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.9.1 Inspection failure. Failure of an inspection comparison tractor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison tractor(s) and shall be cause for refusal by the Government to continue acceptance of production tractors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on tractors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all tractors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.10 Inspection of packaging. Each tractor prepared for shipment shall be inspected for the preservation and packing in accordance with applicable defects contained in MIL-STD-162. In addition, cooling system preservation shall be examined for compliance with this specification. Nonconformance shall be cause for rejection; corrections must be made prior to shipment.

5. PACKAGING

5.1 Preservation and packing. Each complete tractor shall be preserved and packed in accordance with the applicable requirements of MIL-STD-162 for type I mobile. Levels of preservation and of packing shall be A or C as specified (see 6.2). For level C, engine preservation shall be limited to filling the cooling system with a 50/50 mixture of water and antifreeze conforming to MIL-A-46153. The same water/antifreeze mixture shall be used for level A engine preservation in addition to the other required preservation.

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Tractors described herein are intended for towing trailers and equipment in and around warehouses, loading platforms, docks and storage areas within the military supply system. Tractors are intended for operation over paved and semi-paved surfaces.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Size tractor required (see 1.2).
d. When first article test is not specified (see 3.2).
e. Number of tractors in sample for first article test (FAT) (see 3.2).
f. When a GS, DS, or LPS rating is specified (see 3.6).
g. When NBC operability is specified (see 3.7.1).
h. What type of engine is specified (see 3.10).
i. When a closed cab is specified (see 3.15).
j. When a warning light is specified (see 3.15).
k. When treatment, priming, and top coat final color shall be in accordance with MIL-T-704, color 13538 (see 3.18).
l. When CARC coating is specified (see 3.18).
m. When markings shall be painted a color other than black (see 3.19).
n. Level of preservation and of packing required (see 5.1).
o. When a transportability report is required (see 6.5).

6.3 First article. When a first article inspection is required, the items should be a first produced tractor. The first article should consist of 1 or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article. When the contractor can provide verifiable data demonstrating compliance to any or all parts of the specified inspection and testing, these inspections and tests may be waived by the government.

6.4 Incident report. When the contractor conducts the demonstration and test specified herein, a written report within 24 hours, of any malfunction or failure during conduct of the demonstration and test is required. As a minimum, the report shall describe components and parts affected, test and operating conditions, date of incident, hour meter reading, how detected and description of incident and corrective action taken. A minimum of one original will be furnished the pertinent contracting officer specified in the contract. Distribution of copies will be identified in the contract.

6.5 Transportability report. When specified (see 6.2) the contracting officer should arrange for the contractor to furnish a transportability report in accordance with AR 70-47 and S.O.P. 70-38 for the tractor, 95 days prior to delivery of the first article tractor.

6.6 Disposability. One or more of the following methods shall be used to accomplish disposal of lubricating oils, antifreeze, batteries, and paint: reuse, recycling, baling, sanitary landfill, composting, incineration, pyrolysis, or sea disposal.

6.7 Government-loaned property. The contracting officer should arrange to furnish the property listed in 3.7.1.

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6.8 Noise levels. When the required maximum noise limit(s) are established to be technically infeasible and beyond the state-of-the-art for the item/system being procured, the contracting officer may request additional information on noise sources. As such, documentation shall be furnished to the procuring activity for consideration of whether or not increase of the required limit(s) is justified and how noise levels may best be minimized. Documentation shall contain technically defensible data including technically supported design considerations, technically supported design recommendations for noise reduction, and technically supported predictions of the resultant noise levels. Clearly stated, convincing, and technically/fiscally supported trade-off analyses of noise control benefit against other design requirements such as weight, access, etc., shall be provided.

6.9 Subject term (key word) listing.

Tractor, internal combustion machine.
Tractor, pneumatic tired.
Tractor, wheeled.
Tractor, wheeled warehouse.

6.10 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME
Air Force - 99

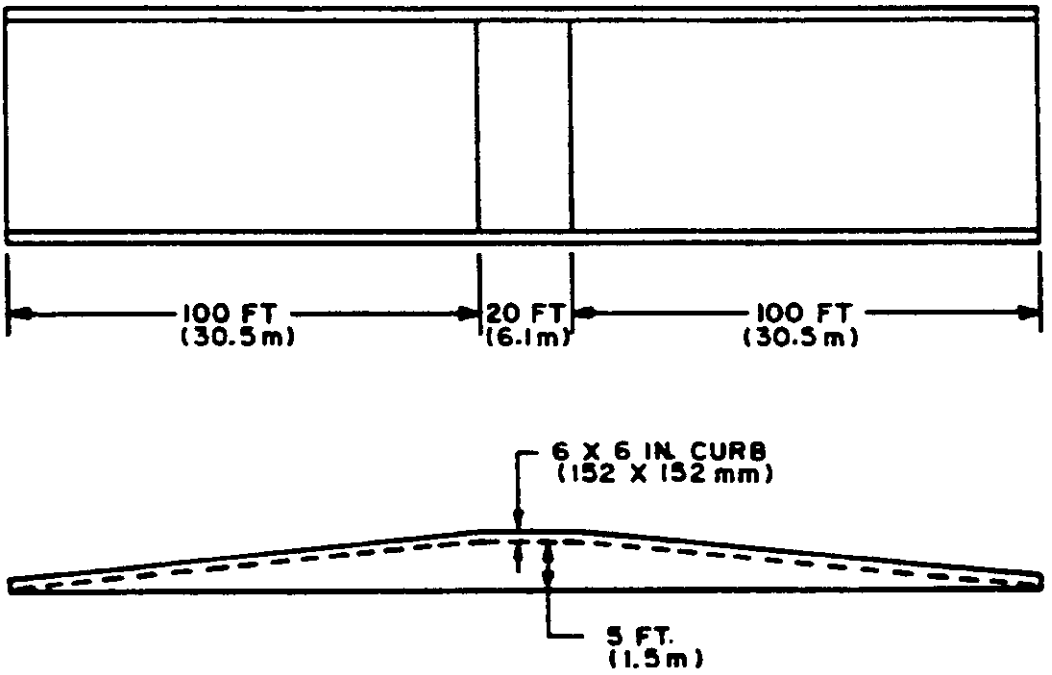
Preparing activity:
Army - ME

Project 3930-0657

Review activities:
Army - SM, MT, AT
Navy - MC
Air Force - 84
DLA - CS

User activity:
Navy - SH, OS, SA

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NOTES:

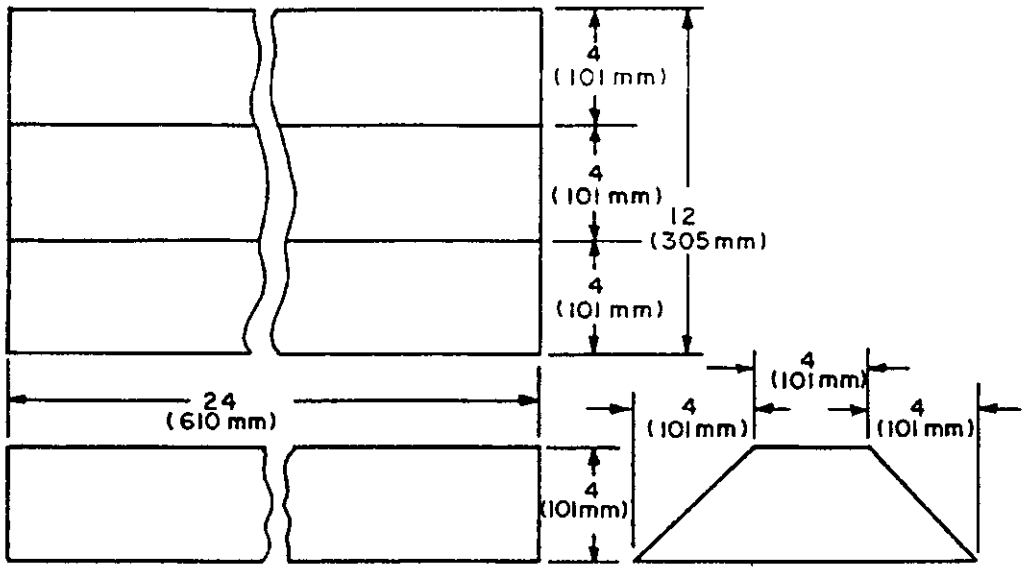
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. RAMP MAY BE ANY WIDTH SUITABLE FOR TRACTOR AND TRAILER.

FIGURE 1. Test ramp.

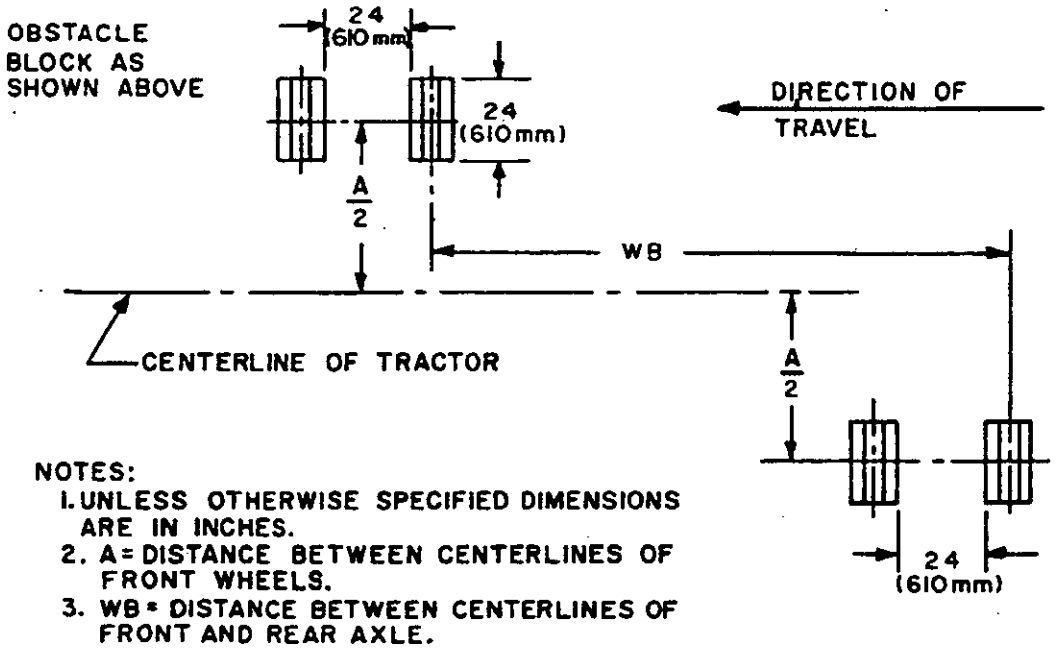
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OBSTACLE BLOCK CONSTRUCTION



LAYOUT OF OBSTACLE COURSE



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. A = DISTANCE BETWEEN CENTERLINES OF FRONT WHEELS.
3. WB = DISTANCE BETWEEN CENTERLINES OF FRONT AND REAR AXLE.

FIGURE 2. Obstacle block layout.

X-4460 A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-52931D	2. DOCUMENT DATE (YYMMDD) 910228

3. DOCUMENT TITLE
Tractor, Wheeled, Warehouse, Internal Combustion Engine, Pneumatic-Tired

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-5717	(2) AUTOVON 354-5717
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:	

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